

SIGNALS IN

OUTCOMES



FRAUDSHIELD AI

"Detect. Investigate. Explain. Act."

Agentic AI for Cyber-Fraud Detection & Explainable Risk Assessment

PROBLEM STATEMENT ID	PROBLEM STATEMENT TITLE	THEME
[PROBLEM STATEMENT ID]	[PROBLEM STATEMENT TITLE]	[THEME]
PS CATEGORY	TEAM ID	TEAM NAME
Software	[TEAM ID]	[TEAM NAME]

Decision-support & early-warning platform — not a replacement for banks, police or official cybercrime systems · Prototype (SIH 2026)



CURRENT EXPERIENCE

A suspicious message arrives — the user is on their own.



Is it real?



Is this link safe?



Should I pay?



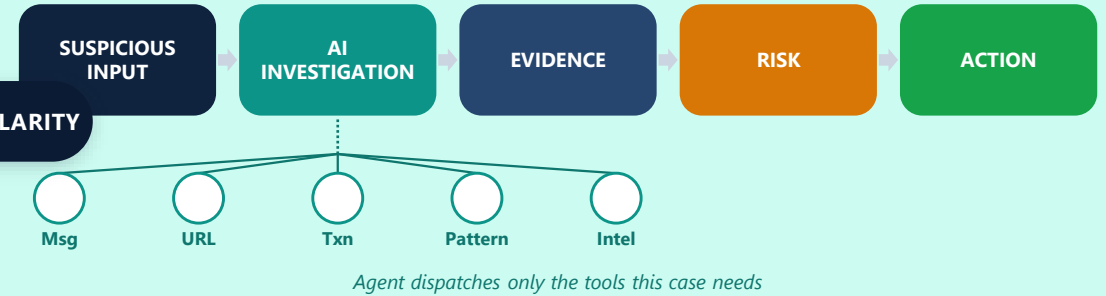
What should I do?



CONFUSION → CLARITY

FRAUDSHIELD AI

One input → multi-signal investigation → explainable outcome.



WHAT MAKES THE AGENT DIFFERENT



AGENTIC

Dynamic tool selection per case



EXPLAINABLE

Every score traces back to visible evidence



ADAPTIVE

Investigation path adapts to each case, not a fixed script

ACTIONABLE

- ✓ Do not click
- ✓ Do not share OTP
- ✓ Verify officially

Core innovation: the agent performs Dynamic Tool Selection instead of running a fixed pipeline.

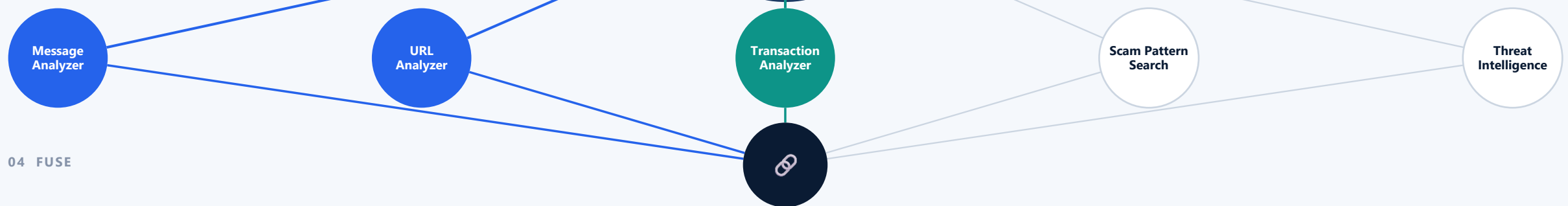
01 INPUT SIGNALS



02 ORCHESTRATOR

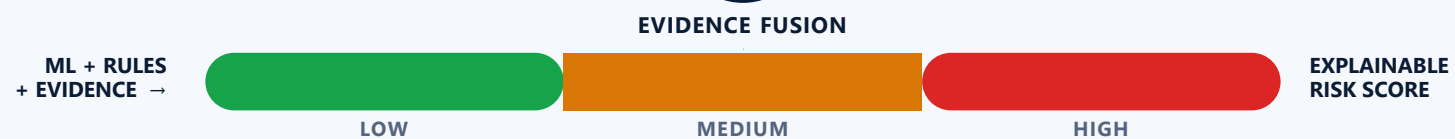


03 TOOL ECOSYSTEM — agent decides which tools to invoke



04 FUSE

05 SCORE



06 DECIDE



07 STORE & TRACK



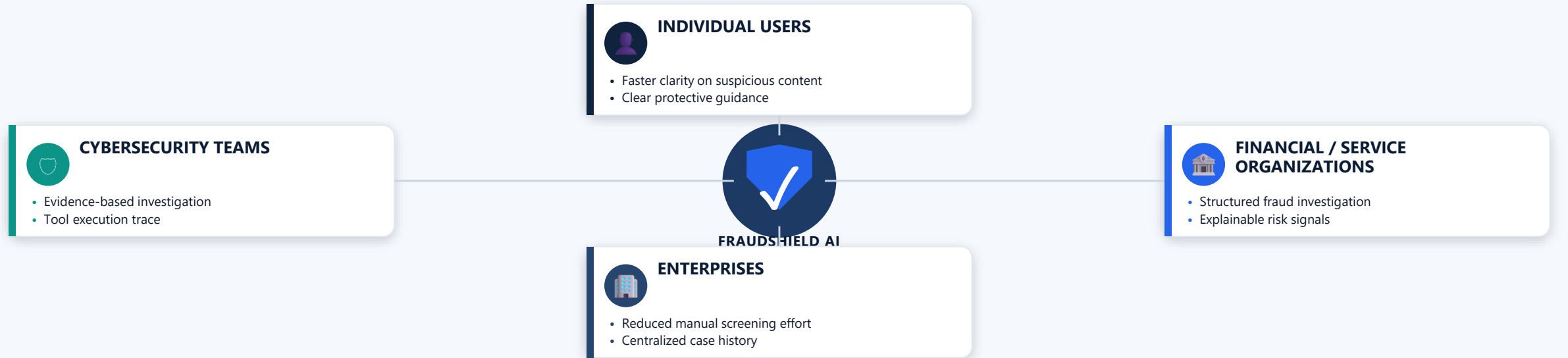


CHALLENGE → MITIGATION CONTROL MAP

! False positives	→	✓ Human review + feedback loop + threshold tuning
! AI hallucination	→	✓ LLM explains structured evidence — never invents it
! Threat-intel limitations	→	✓ Fallback local analysis + configurable adapters
! API / rate limits	→	✓ Caching + graceful fallback
! Sensitive user data	→	✓ Auth + access control + minimal collection

WHY IT IS BUILDABLE

- ✓ Open-source ML
- ✓ Serverless backend
- ✓ Modular agent tools
- ✓ Free-tier prototype stack
- ✓ Incremental deployment



A decision-support & early-warning platform — not a replacement for banks, police or official cybercrime systems.

BEFORE FRAUDSHIELD

Suspicious message

↓

Confusion

↓

Delayed response

↓

Potential exposure

CASE #FS-1024 *Input: Suspicious SMS*

AI INVESTIGATION

- ✓ Input classified
- ✓ URL extracted
- ✓ Message analyzed
- ✓ URL analyzed
- ✓ Scam pattern matched
- ✓ Evidence fused

94%

HIGH RISK

RISK SCORE

WHY

- Suspicious URL
- Urgency language
- Financial request

ACTION

- ✓ Do not click
- ✓ Do not share OTP
- ✓ Verify officially

SAMPLE · ILLUSTRATIVE UI

SOCIAL

Digital safety awareness

SECURITY

Evidence-based investigation

ECONOMIC

Reduced avoidable exposure; streamlined initial screening

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DATA

 - UCI SMS Spam Collection

archive.ics.uci.edu/dataset/228
 - UCI Phishing Websites

archive.ics.uci.edu/dataset/327
 - Credit Card Fraud Detection

Kaggle / ULB dataset
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INTELLIGENCE

 - PhishTank

phishtank.org
 - Threat Intelligence Feeds

configurable adapters (planned)
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TECHNOLOGY

 - LangGraph

langchain-ai.github.io/langgraph
 - Supabase

supabase.com/docs
 - Gemini API

ai.google.dev/gemini-api/docs
 - Scikit-learn

scikit-learn.org

SECURITY REFERENCES

NIST Cybersecurity Guidance

OWASP

CERT-In

Govt. of India Cybercrime Awareness

RESEARCH BASIS

Phishing Detection

SMS Spam Detection

Fraud Detection

Explainable AI

Agentic AI / Tool-Use

— reference to be finalized

PROTOTYPE / DEMO

[ADD WORKING PROTOTYPE LINK]

GITHUB

[ADD GITHUB LINK]

DEMO VIDEO

[ADD VIDEO LINK]